Space-charge distortion of ionization profile monitor measurements in the CSNS rapid-cycling synchrotron

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Abstract
The ionization profile monitor (IPM) of the CSNS rapid-cycling synchrotron records the transverse proton distribution by collecting residual-gas electrons or ions. During their drift to the collector the secondaries are deflected by the bunch space-charge field, so that the detected r.m.s. width differs from the beam size. Particle tracking with Virtual-IPM 2.3.1 is used to quantify that error for the CSNS geometry and to assess the available mitigations. In electron mode a guiding field of 300 G reduces the relative expansion to the one-percent level for all beam families, intensities, cage voltages and orbit offsets examined; the design field of 0.1 T is conservative. In ion mode the same-sign kick always inflates the profile: at 100 kW a 10 mm beam is expanded by 92% (H_2^+, injection) and 104% (H_2^+, extraction). Raising the cage voltage from 5 to 30 kV reduces but does not remove the bias. The residual-gas species is identified in the time-of-flight spectrum and the width of one peak (H_2O^+ or N_2^+) is inverted on a look-up against bunch charge and cage voltage. Because that width has a minimum at a size that grows with intensity (8–19 mm at injection, 6–15 mm at extraction for 100–500 kW), the invert returns two candidates, a core size and a painted-beam size. The guiding magnet is not yet installed, so the electron mode is read at $B = 0$, where its width is itself space-charge biased and no size of its own; it decides between the two candidates by consistency—each candidate predicts a $B = 0$ electron width, and the one that matches the measured electron width is kept—and the second ion species confirms the choice. Tested leave-one-out at 100, 200, 300 and 500 kW at injection (80 MeV) and aligned extraction (1.6 GeV), the electron width picks the right root at every grid point (median margin 34%) and the selected size is within 1% at 92% of them; the 10 mm design core is recovered to $\leq 1\%$ at every power except within one step of the fold, where the electron width is the measurement. At 500 kW injection sizes below 10 mm are lost on the cage. The recommended operating point is electron collection at $B \gtrsim 300$ G.

Keywords
ionization profile monitor; space charge; CSNS RCS; electron collection; beam-size measurement

1 Introduction

An ionization profile monitor infers the transverse beam density from residual-gas ionization products collected on a position-sensitive detector [1, 4]. The CSNS rapid-cycling synchrotron (RCS) IPM [2, 3, 4, 5] uses a 220×231 mm cage biased at a design voltage of 25 kV ($E_y \simeq 108\,\mathrm{kV\,m^{-1}}$) and a magnet specified at 0.1 T (available to 0.2 T). Electrons are collected at the top electrode; the three ion species resolved in the time-of-flight spectrum—H_2^+, H_2O^+ and N_2^+—are collected at the bottom. Two proton bunches circulate. At 100 kW each carries $N = 7.8 \times 10^{12}$ particles (scaled linearly with beam power P). Injection is at 80 MeV ($\beta = 0.39$, $\sigma_t = 120$ ns, painted $\sigma_0 \simeq 25$ mm or a 10 mm core); extraction is at 1.6 GeV ($\beta = 0.93$, $\sigma_t = 20$ ns, $\sigma_0 \simeq 10$ mm).

While the secondaries drift they experience the bunch electric field and the Lorentz-boosted bunch magnetic field. The detected r.m.s. width σ_m is therefore not the beam width σ_0. The same mechanism has been analysed for magnetic electron IPMs by Vilsmeier, Sapinski and Storey [7] and for ion IPMs by Shiltsev [8] and by the ISIS group [9]. The question for CSNS is quantitative: how large is the error, how does it grow with bunch charge and shrink with beam size, and how efficiently do the available mitigations—guiding B, cage voltage, and an identified-species correction—remove it?

The calculation uses Virtual-IPM 2.3.1 [6] with uniform cage fields and 10^5 secondaries per case. Figures 4 and 7–11 are evaluated on 1 kV voltage and 1 mm orbit-offset grids; existing 10^5-particle archives were not re-run. Distortion is isolated by comparing bunch space charge on and off at fixed guiding E and B,

$$\Delta \;=\; \frac{\sigma_{\mathrm{SC}} - \sigma_{\mathrm{off}}}{\sigma_{\mathrm{off}}}\,. \tag{1}$$

A positive Δ is an inflated size; a negative Δ is compression. Microchannel-plate saturation, electromagnetic interference, secondary electrons from the ion trap and the measured cage map are not included. Those hardware effects, rather than space charge, are identified in Refs. [2, 3, 5] as the present electron-mode limit once a sufficient guiding field is applied.

2 Space-charge interaction with collected secondaries

Characteristic times fix the regime. At 25 kV an electron crosses the 115.5 mm half-gap in $\tau \simeq 3.5$ ns. The same path takes $\tau_2 \simeq 210$, 631 and 787 ns for H_2^+, H_2O^+ and N_2^+. The time to leave a 10 mm beam is $\tau_0 \simeq 74$, 221 and 275 ns. Injection bunches (120 ns) are longer than τ_0 for H_2^+; extraction bunches (20 ns) are shorter than every τ_0. The bunch spacing is 980 ns at injection and 409 ns at extraction.

Electrons are attracted to the proton bunch. A moderate kick focuses them towards the axis (compression). When the bunch field exceeds the cage field they cross the axis and the collected profile is wider than the beam (over-focusing). Peak transverse bunch fields on a 10 mm beam are approximately $29\,\mathrm{kV\,m^{-1}}$ (injection) and $73\,\mathrm{kV\,m^{-1}}$ (extraction) at 100 kW, versus 145 and $363\,\mathrm{kV\,m^{-1}}$ at 500 kW; the cage field is $108\,\mathrm{kV\,m^{-1}}$. At 500 kW extraction electrons are trapped in the beam potential during the passage, the regime identified in Ref. [7] as requiring a magnetic field rather than an increase of V.

A guiding field converts the kick into cyclotron motion. If the impulse is delivered early in the flight, the displacement at the collector is proportional to $\sin(\omega_c\tau)/\omega_c$ and vanishes whenever $\omega_c\tau = n\pi$. Successive zeros of $\Delta(B)$ are therefore spaced by

$$\Delta B \;=\; \frac{\pi m_e}{e\tau} \;\propto\; \sqrt{V}\,. \tag{2}$$

The envelope of the oscillation falls as B increases. Magnetic mitigation is the decay of that envelope, not the choice of a particular zero. The same cyclotron-phase picture is the "one gyration" tuning rule used at GSI and J-PARC [10, 11].

Ions have the same sign as the beam, so the kick is always outward and Δ is always positive. In the impulsive limit (bunch length $\ll \tau_0$) the transverse impulse scales as N/σ_0^2 and the subsequent drift as $\tau_2 \propto 1/\sqrt{V}$, giving [8]

$$h - 1 \;\equiv\; \Delta \;\propto\; \frac{N}{V\,\sigma_0^2}\,, \tag{3}$$

with a mass dependence $\propto M^{-1/2}$. CSNS occupies a mixed regime: extraction is impulsive, whereas at injection H_2^+ leaves the beam while the 120 ns bunch is still passing. Magnetic confinement is an electron-mode instrument [7, 8] and is not treated further as an ion-mode correction.

3 Profile distortion and growth of the measured size

3.1 Electron collection without a guiding field

At $B = 0$ the collected electron width is not a reliable size. On the $10\,\mathrm{mm}$ injection beam at $100\,\mathrm{kW}$, Δ changes from expansion to compression at $13\,\mathrm{kV}$ ($+9\%$ at $12\,\mathrm{kV}$, ≈ 0 at $13\,\mathrm{kV}$, -8% at $14\,\mathrm{kV}$). At $500\,\mathrm{kW}$ the same voltage scan never changes sign: Δ remains positive and grows with V from $+29\%$ at $5\,\mathrm{kV}$ to $+83\%$ at $30\,\mathrm{kV}$ (Fig. 1). That is the trapping regime of Section 2: raising V does not un-trap a $500\,\mathrm{kW}$ bunch. The same intensity-dependent sign reversal is reported at GSI (electrons shrink without B at moderate intensity) [10] and at J-PARC (a factor-of-two shrink in electron mode) [11].

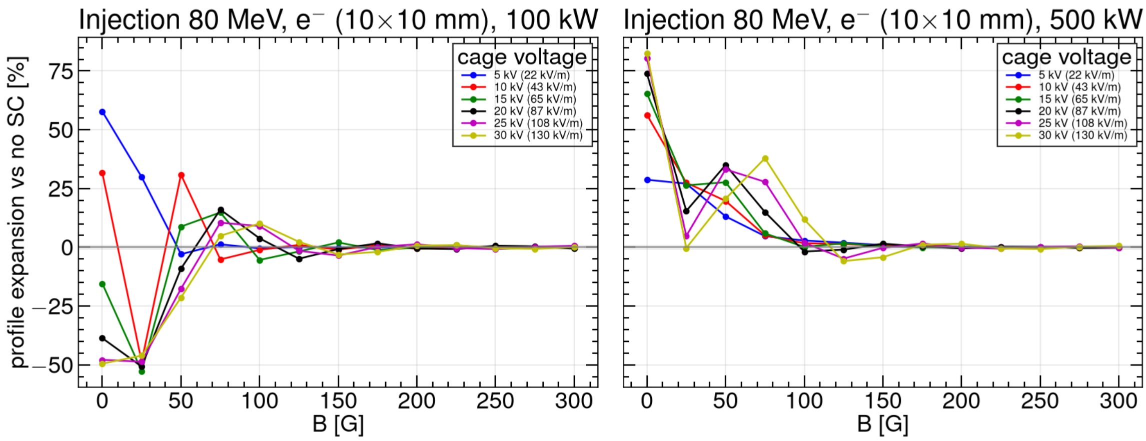

Fig. 1: Relative expansion Δ of the electron profile as a function of guiding field for cage voltages from 5 to $30\,\mathrm{kV}$, on the $10\,\mathrm{mm}$ injection beam. At $B = 0$ the $100\,\mathrm{kW}$ family crosses zero; the $500\,\mathrm{kW}$ family does not.

The map $\sigma_m(\sigma_0)$ is not monotonic at $B = 0$: a small beam may appear larger or smaller than a large one, according to whether the kick focuses or over-focuses. The Voitkiv H_2 double differential cross-section used by Virtual-IPM contributes a σ-independent quadrature smear of 2.9–$3.2\,\mathrm{mm}$ ($\approx$ 2–$2.5\,\mathrm{eV}$ per transverse axis), which is $+40$–48% at $\sigma_0 = 3\,\mathrm{mm}$ and only $+1\%$ at $20\,\mathrm{mm}$. One cyclotron period removes that smear (Section 4.1).

Along the energy ramp at $B = 0$, a long bunch ($120\,\mathrm{ns}$) is compressed and a short bunch ($20\,\mathrm{ns}$) is inflated. Both errors become milder as β rises, because the electrons spend less time in the bunch: at $100\,\mathrm{kW}$ the $120\,\mathrm{ns}$ family goes from -46% ($200\,\mathrm{MeV}$) to -32% ($1.6\,\mathrm{GeV}$), and the $20\,\mathrm{ns}$ family from $+77\%$ ($80\,\mathrm{MeV}$) to $+38\%$ ($1.2\,\mathrm{GeV}$). Energy alone does not restore the profile.

3.2 Ion collection

Every ion case has $\Delta > 0$. Figure 2 shows σ_m versus σ_0 at the design $0.1\,\mathrm{T}$ and $100\,\mathrm{kW}$: every species and both rings lie above the diagonal. Figure 3 shows the same map versus bunch charge: the expansion grows with P and falls with σ_0 as required by Eq. (3), and the observed width lies above $\sigma_m = \sigma_0$ at every power.

For H_2^+ at injection, $\sigma_0 = 10\,\mathrm{mm}$ and $0.1\,\mathrm{T}$, the expansion grows from $+82\%$ ($18.2\,\mathrm{mm}$) at $100\,\mathrm{kW}$ to $+175\%$, $+279\%$, $+390\%$ and $+422\%$ ($52.1\,\mathrm{mm}$) at 200, 300, 400 and $500\,\mathrm{kW}$. Between 100 and $400\,\mathrm{kW}$ the growth is $\propto P^{1.12}$, consistent with the linear kick plus the second-order term of $\sigma_m^2 = \sigma_0^2 + \kappa N + \cdots$ [8]. At $500\,\mathrm{kW}$ the profile reaches the $110\,\mathrm{mm}$ half-cage and the exponent saturates.

At extraction, with ions born in time with the bunch, $10\,\mathrm{mm}$, $100\,\mathrm{kW}$ and $25\,\mathrm{kV}$, the expansions are $+104\%$ (H_2^+), $+41\%$ (H_2O^+) and $+40\%$ (N_2^+) at $B = 0$, in agreement with a reduced line-charge

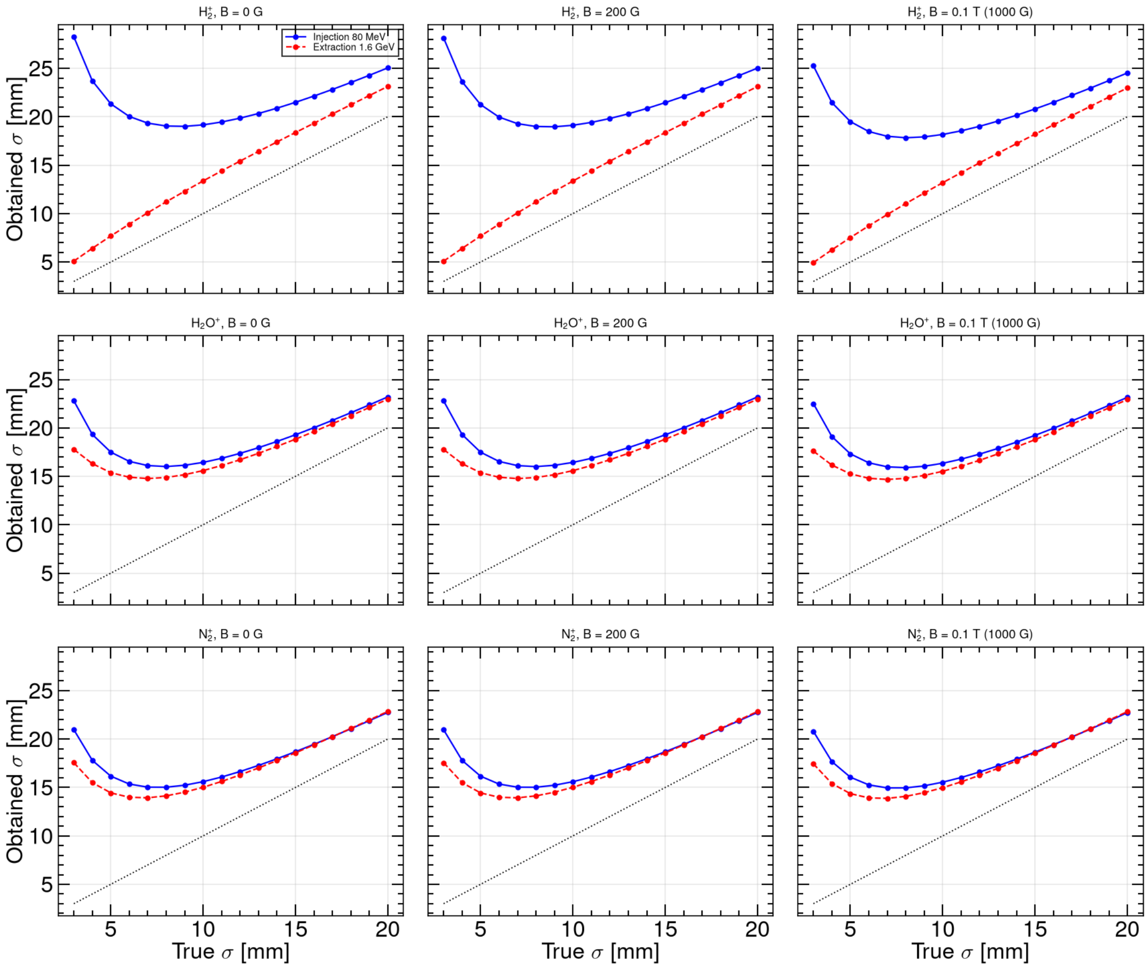

Fig. 2: Ion-mode obtained width σ_m versus true width σ_0 at $0.1\,\mathrm{T}$ and $100\,\mathrm{kW}$. The dashed line is $\sigma_m = \sigma_0$. Space charge inflates every point.

integration of the same equations to $\pm 1\%$ absolute (Fig. 4; model values $+102\%$, $+40\%$ and $+39\%$). The $20\,\mathrm{ns}$ bunch is impulsive, so H_2^+ is the most distorted species. At $500\,\mathrm{kW}$ the same family reaches $+408\%$, $+243\%$ and $+235\%$. Between $\sigma_0 = 5$ and $20\,\mathrm{mm}$ at $100\,\mathrm{kW}$, $\Delta \propto \sigma_0^{-1.84\ldots-2.01}$, the impulsive σ_0^{-2} of Eq. (3) rather than the continuous-beam $\sigma_0^{-1.5}$. Aligned extraction H_2^+ widths at $0\,\mathrm{G}$ and $100\,\mathrm{kW}$ are listed in Table 1. A $3\,\mathrm{mm}$ extraction beam is already unusable as a size measurement at $100\,\mathrm{kW}$. The most severe case remains injection at $3\,\mathrm{mm}$ and $500\,\mathrm{kW}$, where $\sigma_m \approx 56\,\mathrm{mm}$ ($+1700\%$) for all three species.

Table 1: Aligned extraction, H_2^+, $0\,\mathrm{G}$, $100\,\mathrm{kW}$, $25\,\mathrm{kV}$: obtained width and expansion versus true size.

σ_0 (mm)	3	5	7	10	15	20
σ_m (mm)	38.2	26.8	22.4	20.4	21.7	24.9
Δ (%)	+1172	+435	+219	+104	+44	+24

Species ordering follows the mixed regime. At injection, $10\,\mathrm{mm}$, $0\,\mathrm{G}$ and $100\,\mathrm{kW}$ the expansions are $+92\%$ (H_2^+), $+65\%$ (H_2O^+) and $+56\%$ (N_2^+): lighter ions expand more, but less than a pure $M^{-1/2}$ law because H_2^+ leaves during the $120\,\mathrm{ns}$ bunch. Painted injection ($25\,\mathrm{mm}$) is milder ($+17\%$, $+10\%$, $+9\%$). A second extraction bunch, still in the cage when the heavy ions arrive, adds a few points of extra

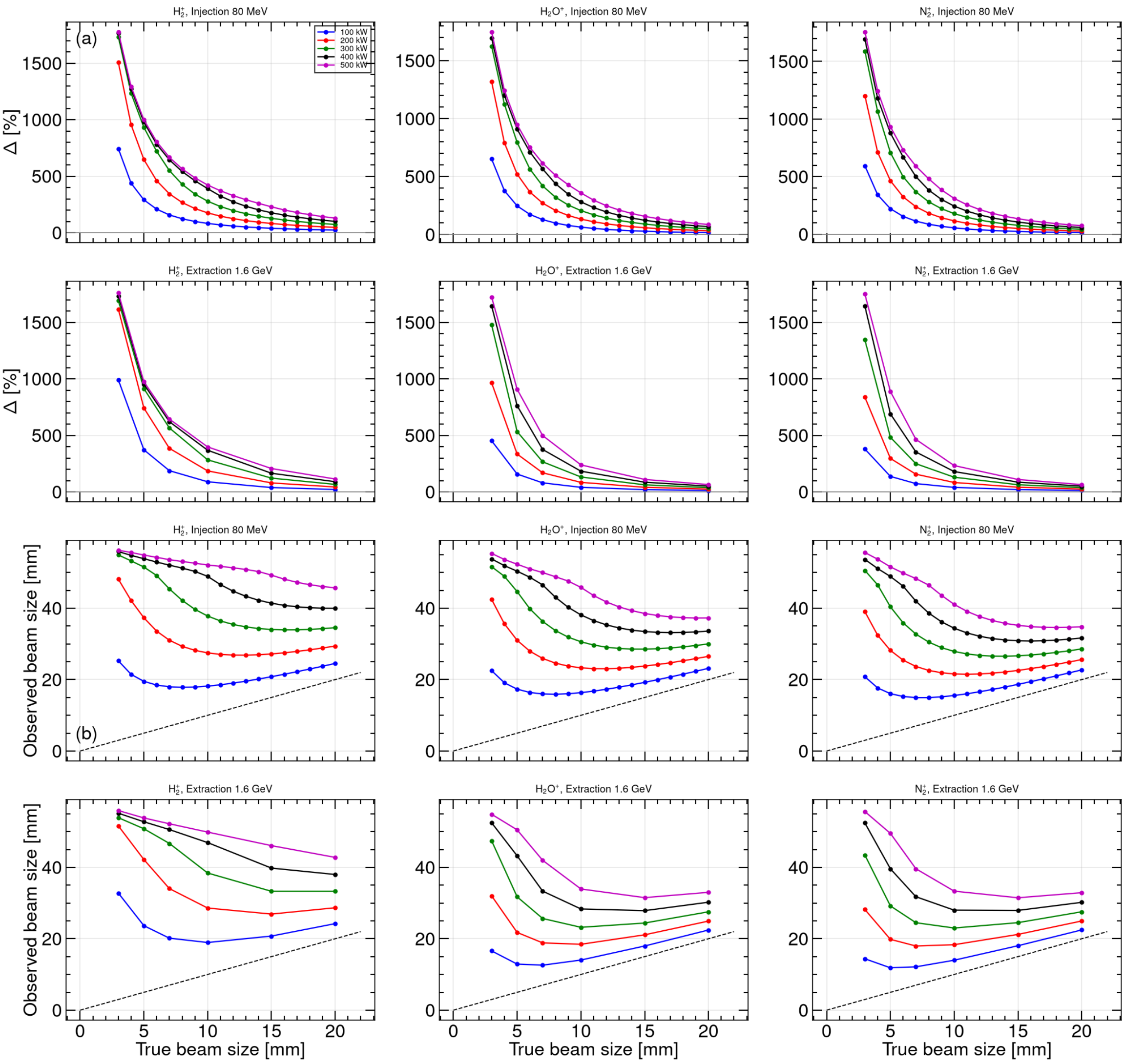

Fig. 3: (a) Ion-mode expansion Δ versus true size σ_0 at $0.1\,\mathrm{T}$ for 100–$500\,\mathrm{kW}$. Injection is the Block B size scan; extraction uses the aligned bunch timing (ion born with the bunch). (b) Observed width σ_m versus true width σ_0 for the same points. The dashed line is $\sigma_m = \sigma_0$. The $3\,\mathrm{mm}/500\,\mathrm{kW}$ family saturates on the cage wall ($\sigma_m \approx 56\,\mathrm{mm}$).

width (single-bunch $+36\%$ and $+29\%$ versus three-bunch $+41\%$ and $+40\%$ for H_2O^+ and N_2^+); H_2^+ does not see that bunch ($+104\%$ either way). At commissioning power, aligned extraction H_2^+ ($10\,\mathrm{mm}$, $0\,\mathrm{G}$) is already $+19\%$ at $20\,\mathrm{kW}$ and $+82\%$ at $80\,\mathrm{kW}$. These magnitudes are of the same order as the operational ion-mode biases reported at J-PARC RCS (20–30% wider than the MWPM) [11] and at the Fermilab Booster [8].

4 Mitigation

4.1 Magnetic confinement of electrons

Figure 5 shows the electron expansion versus B for the three reference beams and 100–$500\,\mathrm{kW}$. The curves oscillate through focusing and over-kick. Painted injection recovers first; extraction and the $10\,\mathrm{mm}$ injection core still oscillate through 150–$250\,\mathrm{G}$. At $300\,\mathrm{G}$ every family has settled inside $\pm 1\%$ (Fig. 6).

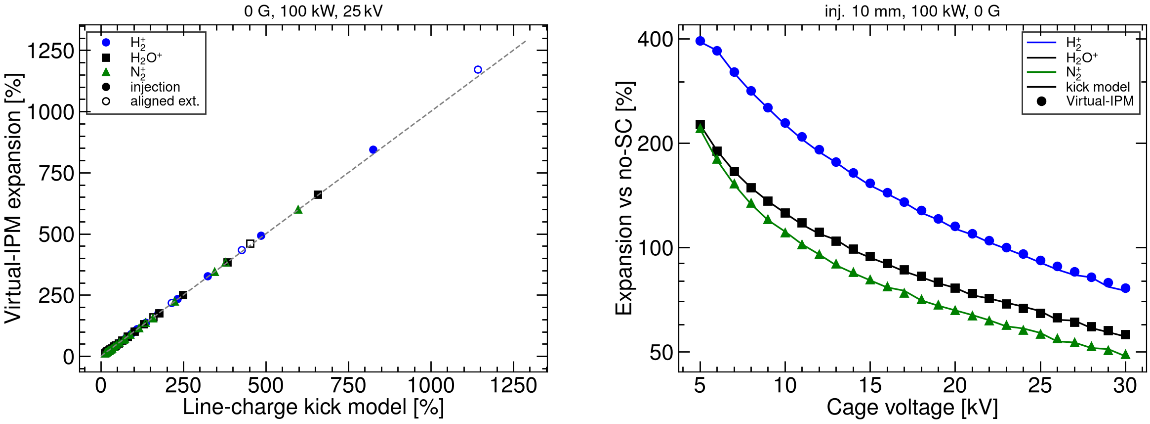

Fig. 4: Left: ion-mode expansion at 0 G, 100 kW and 25 kV, reduced kick model versus Virtual-IPM. Filled markers: injection $\sigma_0 = 3$–$20\,\mathrm{mm}$. Open markers: aligned extraction (3, 5, 7, 10, 15, 20 mm). Right: cage-voltage scan at 1 kV steps, injection 10 mm. Curves: kick model. Markers: Virtual-IPM. Agreement is $\pm 1\%$ absolute on the left and $\leq 4\%$ relative on the right.

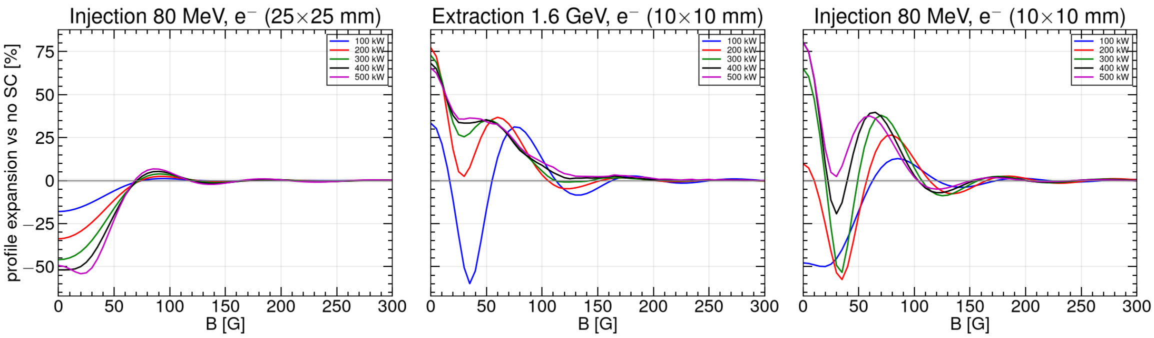

Fig. 5: Electron-mode expansion versus guiding field, 0–300 G, for painted injection (25 mm), extraction (10 mm) and the 10 mm injection core, at 100–500 kW. The grey band is $\pm 1\%$.

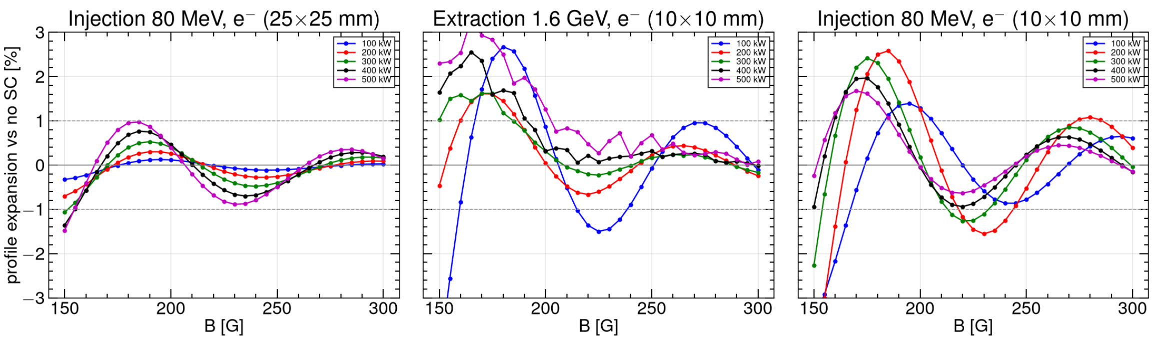

Fig. 6: As Fig. 5, 150–300 G, $\pm 3\%$ vertical scale. At 300 G every family lies inside $\pm 1\%$.

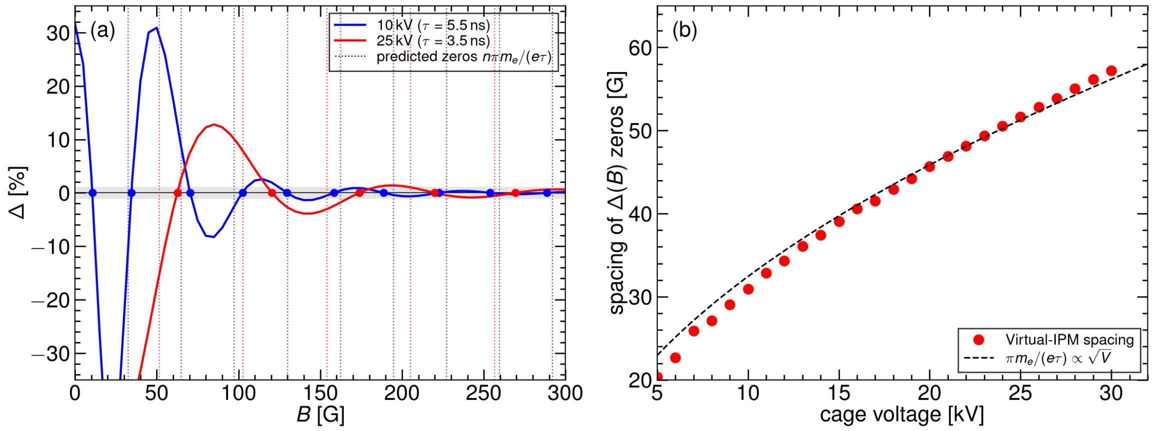

Fig. 7: (a) Relative expansion Δ versus guiding field at two cage voltages (100 kW, 10 mm injection). Dotted vertical lines (colour-matched to the curves) are the predicted zeros $B_n = n\pi m_e/(e\tau)$; circles are the simulated crossings. The grey band is $\pm 1\%$. The frame is $\pm 35\%$; the 25 kV point at $B = 0$ ($\Delta = -48\%$) remains off-scale. (b) Mean spacing of those zeros versus cage voltage at 1 kV steps, compared with $\pi m_e/(e\tau) \propto \sqrt{V}$ (no free parameter).

Figures 5 and 6 do not fall monotonically: $\Delta(B)$ oscillates through focusing ($\Delta < 0$) and overkick ($\Delta > 0$). Figure 7 identifies that oscillation as the cyclotron phase of a nearly impulsive space-charge kick.

An electron born on a 10 mm beam leaves the bunch in $\lesssim 1$ ns and then drifts the remaining $\tau \simeq 3.5$ ns (25 kV) in the uniform cage field. The transverse impulse Δv acquired inside the bunch is converted into cyclotron motion at $\omega_c = eB/m_e$. The displacement at the collector is $(\Delta v/\omega_c)\sin(\omega_c\tau)$ and therefore vanishes whenever the electron completes an integer number of half-turns, $\omega_c\tau = n\pi$ ($n = 1, 2, \ldots$). Those fields are exactly the equally spaced zeros $B_n = n\,\pi m_e/(e\tau)$ of Eq. (2). Because $\tau \propto V^{-1/2}$ in a uniform field, the spacing itself scales as $\sqrt{V}$. The amplitude of each successive lobe falls as $1/B$ (the $1/\omega_c$ prefactor). Magnetic mitigation is the decay of that envelope, not the choice of a particular zero: a zero can be moved by a few kilovolts or a change of orbit, whereas the envelope is a property of B.

Panel (a) is a 5 G-step scan at 100 kW and 10 mm injection, drawn at 10 and 25 kV, with vertical range $\pm 35\%$. Vertical dotted lines are the predicted zeros $B_n = n\pi m_e/(e\tau)$ for each voltage; filled circles mark the Virtual-IPM crossings. The 10 kV curve oscillates faster because the flight is longer ($\tau = 5.51$ ns versus 3.48 ns), so a given phase $n\pi$ is reached at a weaker B. The first 25 kV point, $\Delta = -48\%$ at $B = 0$, lies below the frame. At 25 kV the remaining lobes sit near 85, 140, 195, 240 and 295 G with amplitudes 13, 4, 1.4, 0.9 and 0.6%.

Panel (b) extracts the mean spacing between successive zeros on the 5–30 kV grid at 1 kV steps (Table 2 lists every voltage). The points follow the $\sqrt{V}$ curve to $\leq 5\%$ on the 5 G C2 voltages; the 10 G C3 fill at 5–8 kV is coarser than a half-lobe and sits 5–11% low. There is no free parameter. That is the evidence that the oscillation in Figs. 5 and 6 is the cyclotron phase, and it is why 300 G—the first field at which the envelope itself lies inside $\pm 1\%$ at every power—is the operating point rather than any particular zero.

Table 3 lists the smallest field after which $|\Delta|$ remains below 1%, together with the residual at 300 G. Painted injection is restored first. The last curve to settle is the 10 mm injection core at 200 kW (290 G). A field of 300 G keeps every power at $|\Delta| \lesssim 1\%$.

The same 300 G point holds when the remaining axes are varied. From 200 G the obtained-versus-

Table 2: Electron time of flight from the cage mid-plane and the spacing of $\Delta(B)$ zeros on the 1 kV grid (injection 10 mm, 100 kW). The prediction is Eq. (2) with $\tau = \sqrt{2dm_e/(eE)}$ and $d = 115.5$ mm. C2 voltages (10, 12, 15, 18, 20, 25 kV) use a 5 G B scan; the others are the 10 G C3 fill.

V (kV)	τ (ns)	pred. (G)	sim. (G)	step (G)	V (kV)	τ (ns)	pred. (G)	sim. (G)	step (G)
5	7.79	22.9	20.3	10	18	4.11	43.5	42.9	5
6	7.11	25.1	22.7	10	19	4.00	44.7	44.2	10
7	6.58	27.1	25.9	10	20	3.89	45.9	45.7	5
8	6.16	29.0	27.1	10	21	3.80	47.0	46.9	10
9	5.81	30.8	29.0	10	22	3.71	48.1	48.2	10
10	5.51	32.4	30.9	5	23	3.63	49.2	49.4	10
11	5.25	34.0	32.9	10	24	3.56	50.2	50.5	10
12	5.03	35.5	34.3	5	25	3.48	51.3	51.7	5
13	4.83	37.0	36.1	10	26	3.42	52.3	52.8	10
14	4.66	38.4	37.4	10	27	3.35	53.3	53.9	10
15	4.50	39.7	39.1	5	28	3.29	54.3	55.0	10
16	4.35	41.0	40.6	10	29	3.23	55.2	56.2	10
17	4.22	42.3	41.5	10	30	3.18	56.2	57.2	10

Table 3: Smallest guiding field after which $|\Delta| \leq 1\%$ for the remainder of the 0–300 G grid. The residual at 300 G is given in parentheses.

Beam	100 kW	200 kW	300 kW	400 kW	500 kW
Inj. 25 mm	105 G (+0.03%)	115 G (+0.09%)	155 G (+0.16%)	155 G (+0.19%)	155 G (+0.15%)
Ext. 10 mm	240 G (−0.11%)	190 G (−0.24%)	185 G (−0.17%)	195 G (−0.02%)	205 G (+0.08%)
Inj. 10 mm	210 G (+0.61%)	290 G (+0.40%)	235 G (−0.04%)	190 G (−0.16%)	190 G (−0.15%)

true plot lies on the diagonal except at 3 mm (Fig. 8). At 300 G, injection at 3 mm is still $+1\%$ (100 kW) to $+4\%$ (500 kW); 0.1 T places 3, 10 and 20 mm on the diagonal at every power ($|\Delta| \leq 0.05\%$). At 300 G and 10–25 kV, both 100 and 500 kW remain $\lesssim 1\%$. Higher V requires slightly more B at 100 kW (150 G at 10 kV to 210 G at 25 kV) because τ is shorter and the first lobes move out; 500 kW recovers earlier at every voltage. Along the energy ramp, 300 G and 100 kW give $|\Delta| \leq 0.5\%$ from 80 MeV to 1.6 GeV for both 120 ns and 20 ns bunches. At commissioning powers (20/50/80 kW) and 300 G the residuals are +0.00/+0.01/+0.02% (painted injection), +0.28/+0.56/+0.18% (extraction 10 mm) and +0.11/+0.29/+0.49% (injection 10 mm).

A few millimetres of vertical orbit offset change Δ by a few percent at $B = 0$ and by $\lesssim 1\%$ at 300 G; the 1% threshold moves by at most ~ 10 G. A horizontal offset is a translation of a uniform cage and does not change the width. The space-charge-induced centroid shift falls as $1/B$ (the $\mathbf{E} \times \mathbf{B}$ / polarisation drift of Ref. [7]): 0.9 mm at 100 G to 0.2 mm at 300 G on the extraction beam.

Thus $B = 300$ G reduces an electron-mode error of tens of percent (and the wrong sign at 100 kW and high V) to $\lesssim 1\%$. The design 0.1 T provides a factor-of-three margin and is the appropriate field for a 3 mm beam. The CSNS magnet (0.2 T) is not the limiting device. The SNS ring IPM was designed for 300 G at twenty-five times this bunch charge, with an estimated residual of about 7% [12]; a residual $\lesssim 1\%$ at CSNS is the expected scaling. The CERN PS beam-gas ionization monitor operates at 200 mT for LHC-type bunches [13], at the high-field end of the same scaling.

4.2 Cage voltage

For electrons at $B = 0$, voltage is not a size correction: it can reverse the sign of Δ at 100 kW and cannot reverse it at 500 kW (Section 3.1). Once $B \geq 300$ G the design 25 kV already lies inside 1%, so the voltage may be chosen for collection efficiency and detector operation.

For ions, a higher V shortens τ_2 and reduces the integrated kick, as at ISIS (broadening $\propto$

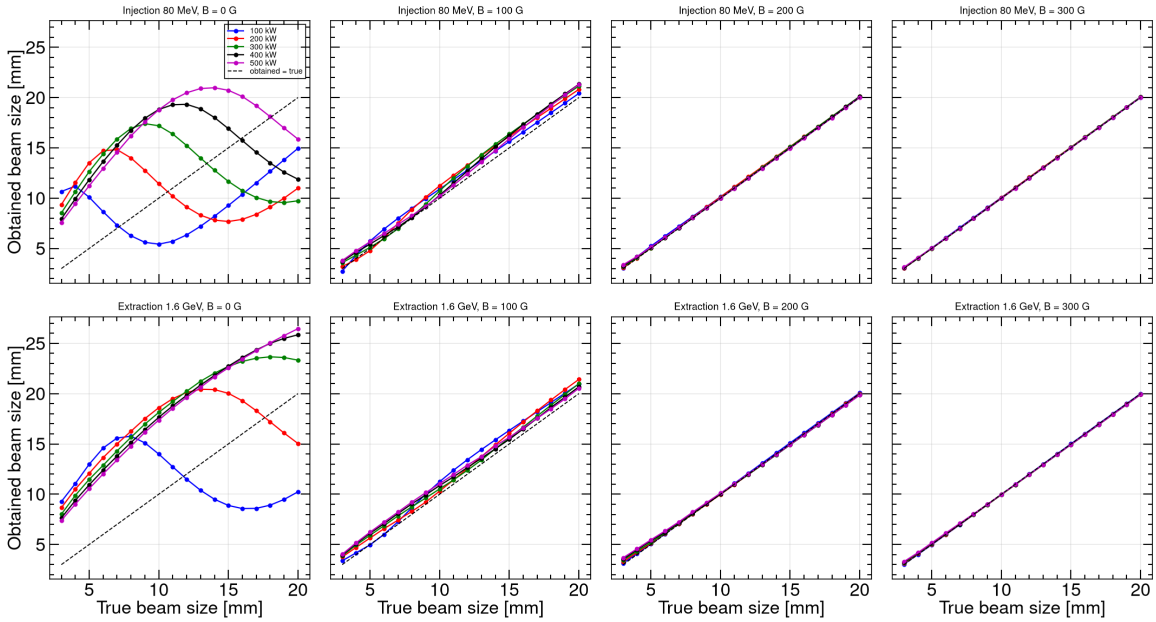

Fig. 8: Electron-mode obtained width versus true width at $B = 0$, 100, 200 and 300 G. From 200 G the points lie on the diagonal except at $3\,\mathrm{mm}$.

$1/E$) [9]. The relevant ion-mode scan is therefore $\Delta(V)$, not $\Delta(B)$: even at the design $0.1\,\mathrm{T}$ the ion cyclotron phase is $\omega_c \tau_2 \lesssim 1\,\mathrm{rad}$, so a guiding field is not a size correction.

Figure 9 is that voltage scan for the three time-of-flight-resolved species on the $10\,\mathrm{mm}$ injection beam at $100\,\mathrm{kW}$ and $B = 0$. Curves are the line-charge kick model at $1\,\mathrm{kV}$ steps from 5 to $30\,\mathrm{kV}$; markers are Virtual-IPM on the same $1\,\mathrm{kV}$ grid. The model tracks every marker to $\leq 4\%$ relative (≤ 4 points absolute). Expansion remains positive at every voltage: from 5 to $30\,\mathrm{kV}$, H_2^+ falls from $+395\%$ to $+77\%$, H_2O^+ from $+227\%$ to $+56\%$, and N_2^+ from $+221\%$ to $+49\%$ (Table 4). The Virtual-IPM points on the $1\,\mathrm{kV}$ grid scale as $V^{-0.96}$ (H_2^+), $V^{-0.75}$ (H_2O^+) and $V^{-0.80}$ (N_2^+). Aligned extraction H_2^+ ($10\,\mathrm{mm}$, $100\,\mathrm{kW}$) falls from $+280\%$ at $5\,\mathrm{kV}$ to $+104\%$ at $25\,\mathrm{kV}$ and $+93\%$ at $30\,\mathrm{kV}$. Extrapolating $V^{-0.96}$ from the $25\,\mathrm{kV}$ injection point, a 10% bias would require approximately $250\,\mathrm{kV}$ across the cage, which is not a practical path. ESS can operate an ion IPM at $300\,\mathrm{kV\,m^{-1}}$ because its bunches are weaker by about 5000 [14]; that choice does not transfer to CSNS.

Table 4: Ion-mode expansion versus cage voltage at $B = 0$ (injection $10\,\mathrm{mm}$, $100\,\mathrm{kW}$). Virtual-IPM on the $5\,\mathrm{kV}$ subset of the $1\,\mathrm{kV}$ scan.

V (kV)	5	10	15	20	25	30
H_2^+	$+395$	$+229$	$+153$	$+115$	$+92$	$+77$
H_2O^+	$+227$	$+126$	$+94$	$+77$	$+65$	$+56$
N_2^+	$+221$	$+110$	$+81$	$+66$	$+56$	$+49$

Voltage is therefore a useful lever on the ion bias (a factor of about five for H_2^+ from 5 to $30\,\mathrm{kV}$) but not a substitute for correction or for electron-mode operation.

4.3 Closed-orbit offset

A few millimetres of closed-orbit error are routine in the RCS. The space-charge kick is centred on the beam, so a horizontal offset Δx is a translation of a uniform cage and must leave the measured width unchanged. A vertical offset Δy changes the drift length $d = 115.5\,\mathrm{mm} + \Delta y$ and therefore the flight

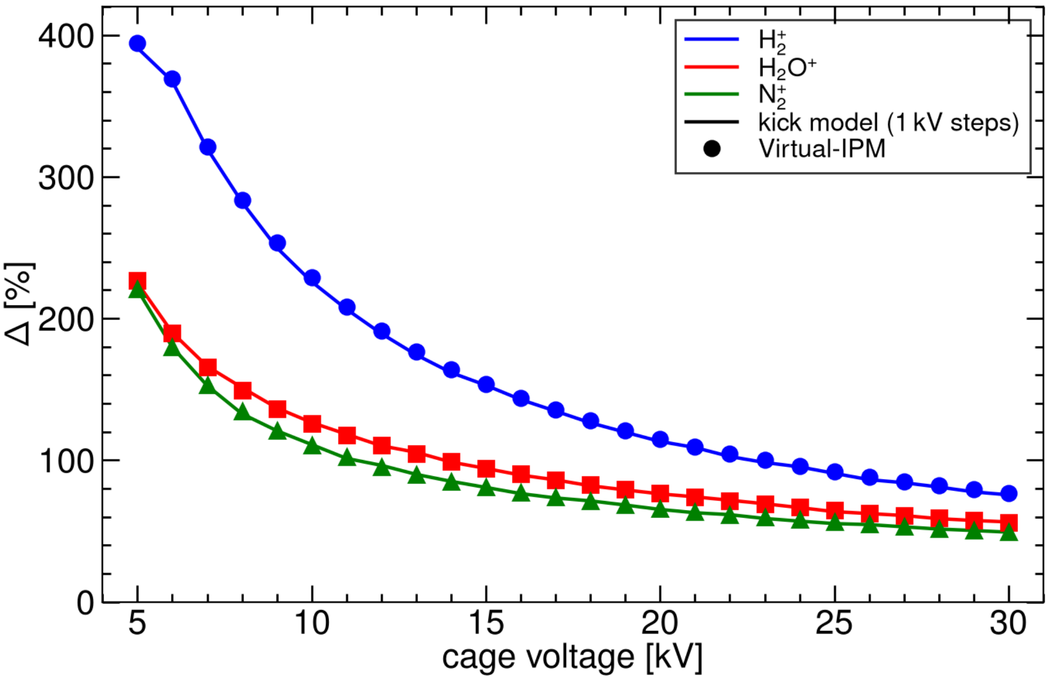

Fig. 9: Ion-mode expansion versus cage voltage for H_2^+, H_2O^+ and N_2^+ (injection $10\,\mathrm{mm}$, $100\,\mathrm{kW}$, $B = 0$). Curves: kick model at $1\,\mathrm{kV}$ steps from 5 to $30\,\mathrm{kV}$. Markers: Virtual-IPM on the same grid. Guiding B is not scanned.

time; that is the only first-order geometric lever.

Figure 10 is the electron check on the $10\,\mathrm{mm}$ beams at $100\,\mathrm{kW}$. Panel (a) is the Fine D scan $\Delta y = \pm 10\,\mathrm{mm}$ in $1\,\mathrm{mm}$ steps. At $B = 0$ extraction expands from $+25\%$ ($\Delta y = -10\,\mathrm{mm}$) to $+42\%$ ($\Delta y = +10\,\mathrm{mm}$), 0.85% per millimetre, because the electron stays longer in the bunch field when the beam is farther from the collector; injection at $B = 0$ is compressed ($\Delta \simeq -48\%$) and almost flat. At $300\,\mathrm{G}$ both beams lie inside the $\pm 1\%$ band over the whole $\pm 10\,\mathrm{mm}$ range ($|\Delta| \leq 0.65\%$). Panel (b) is the horizontal scan at that operating field in $1\,\mathrm{mm}$ steps from 0 to $10\,\mathrm{mm}$: Δ is unchanged to the line thickness (injection $+0.61\%$, extraction -0.11% at the centred point). The $\mathbf{E} \times \mathbf{B}$ centroid shift ($0.24\,\mathrm{mm}$ on extraction at $300\,\mathrm{G}$) is independent of Δx and is a fixed, correctable offset.

Figure 11 is the ion check at $B = 0$ (a guiding field is not an ion-mode lever). Panel (a) plots Δ versus Δy for the three species. Filled markers are injection; open markers are aligned extraction. Lines are $\sigma_m^2 - \sigma_0^2 \propto d$, anchored at the centred Virtual-IPM point, with no free parameter. Injection H_2^+ is $+89.2\%$, $+92.0\%$ and $+94.7\%$ at $\Delta y = -5$, 0 and $+5\,\mathrm{mm}$; the prediction is $+88.9\%$ and $+95.0\%$. Aligned extraction H_2^+ is $+101\%$, $+104\%$ and $+107\%$ on the same Δy grid. The $1\,\mathrm{mm}$ scan from -5 to $+5\,\mathrm{mm}$ stays on that line. Every species and both beams agree with the drift-length law to 0.5% absolute. Panel (b): the $1\,\mathrm{mm}$ Δx scan from 0 to $10\,\mathrm{mm}$ leaves every width unchanged to 0.1%, as required by a kick centred on the beam and as stated for ISIS within about $10\,\mathrm{mm}$ of the axis [9].

At $500\,\mathrm{kW}$ an uncorrected injection H_2^+ profile ($\sigma_m \approx 52\,\mathrm{mm}$) is clipped by the $110\,\mathrm{mm}$ half-cage. A $10\,\mathrm{mm}$ horizontal offset then pulls the centroid by $4\,\mathrm{mm}$. That is an aperture effect, not a failure of the electron-mode operating point and not a space-charge shift of the ion image.

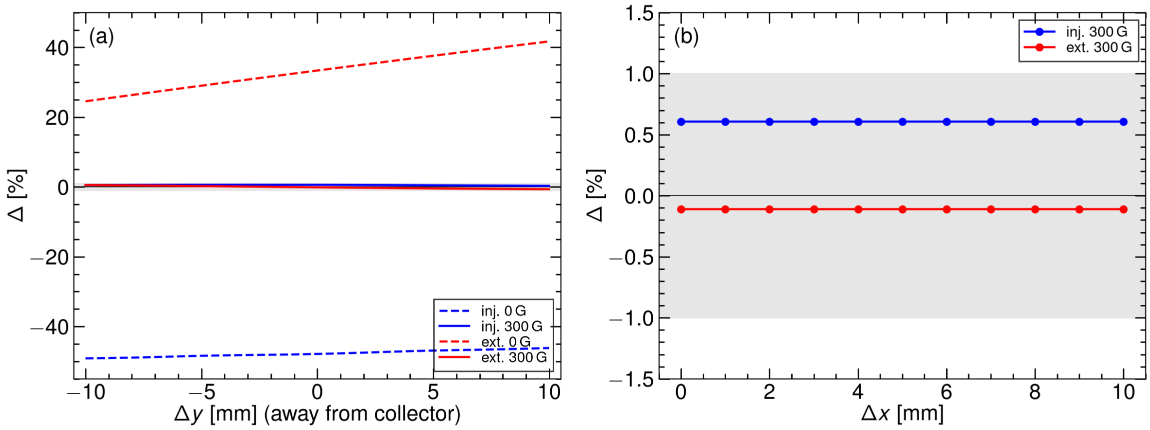

Fig. 10: Electron-mode closed-orbit offset, 10 mm beams, 100 kW. (a) Δ versus Δy (Fine D, 1 mm steps). Dashed: $B = 0$. Solid: 300 G. The grey band is $\pm 1\%$. (b) Δ versus Δx at 300 G (0–10 mm, 1 mm steps).

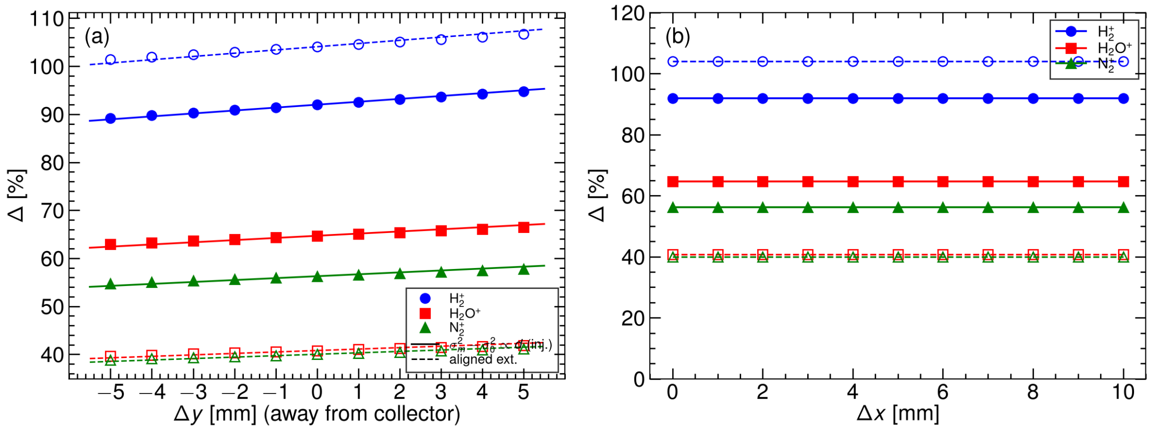

Fig. 11: Ion-mode closed-orbit offset at $B = 0$, 10 mm, 100 kW, 25 kV. (a) Δ versus Δy (-5 to $+5$ mm, 1 mm steps). Filled: injection. Open: aligned extraction. Lines: $\sigma_m^2 - \sigma_0^2 \propto d$. (b) Δ versus Δx (0–10 mm, 1 mm steps); the width is translation-invariant. Guiding B is not scanned.

4.4 Inversion of ion-mode widths

The ion bias is large, smooth, and reproduced by the kick model (Fig. 4), so σ_0 can be recovered from σ_m without changing the hardware. The residual-gas species is identified from the time-of-flight spectrum, and the working peak is a single identified species—H_2O^+ or N_2^+, which are less impulsive than H_2^+ (Section 2). Its look-up $\sigma_m(\sigma_0, N, V)$ at $B = 0$ (Virtual-IPM Block B at injection, aligned I2 at extraction; 1 mm steps, 3–20 mm, 25 kV) is then inverted. The guiding magnet is not yet installed, so both the ion mode and the electron mode are read at $B = 0$ throughout this section.

Two roots. The inversion is not single-valued. At fixed N and V the collected width of a heavy ion has a minimum $\sigma_{m,\min}$ at $\sigma_0 = \sigma_0^*(N)$: below it the kick term ($\propto N/\sigma_0$) dominates and σ_m *falls* as the beam grows; above it the beam size dominates and σ_m rises. A measured width above $\sigma_{m,\min}$ therefore has two roots, $\sigma_S \le \sigma_0^* \le \sigma_L$. σ_S is the *core* hypothesis (a small, dense beam whose ions are blown out), σ_L the *painted-beam* hypothesis (a large beam whose ions are barely displaced). The fold moves out with intensity, roughly as $N^{1/2}$ (Block B H_2O^+: 8, 12, 15 and 19 mm at 100, 200, 300 and 500 kW; aligned extraction: 6, 9, 11 and 15 mm), so the same 10 mm design core is the large root at 100 kW and the small root at 300 kW. The closed-form Shiltsev law $\sigma_m = \sigma_0 + \kappa/\sigma_0$ [8] has the same two roots but not the shape of the minimum; each branch of the CSNS table is therefore interpolated separately with a monotone cubic.

Choosing the root. Both roots are returned, and the electron mode decides between them. Without a guiding field the electron width σ_e is not a size of its own: the electrons are focused or over-focused in the bunch potential during their drift, and $\sigma_e(\sigma_0)$ swings from -48% (10 mm, 100 kW injection) to $+80\%$ (500 kW) and is itself non-monotonic (Section 3.1; black curves in Figs. 12 and 13). It is, however, a deterministic function of the same (σ_0, N, V), tabulated by the same Virtual-IPM runs. The check is therefore one of consistency: each ion root predicts an electron width, $T_e(\sigma_S)$ and $T_e(\sigma_L)$, from the $B = 0$ electron table at the measured N and V, and the root whose prediction matches the measured σ_e is kept. The discriminating power is the margin $|T_e(\sigma_S) - T_e(\sigma_L)|/\sigma_e$: it is 34% in the median over the grid, above 9% at 90% of the points, and below 5% at six of 165 two-root points—five of them within one step of the fold, where the two roots are anyway within a few percent of each other, and one at 17 mm, 100 kW injection (H_2O^+, 3.7%; N_2^+ gives 6% there). A measured electron width good to a few percent therefore settles the choice everywhere it matters. The second identified species is an independent check that needs no electron data: the correct roots of H_2O^+ and N_2^+ agree to $\lesssim 0.1\%$ (median; 1% on the two-point 100 kW extraction core branch), while their wrong roots disagree by 10–15% at injection and 14–30% at 100–300 kW extraction, because a wrong root scales with the kick strength of the species. Only at 500 kW extraction, where the two heavy-ion curves nearly coincide, does that separation shrink to 3–4%, and the electron check carries the decision. Once the magnet is installed, σ_e at $B \geq 300$ G is within one percent of σ_0 (Section 4.1) and the check reduces to keeping the root nearer to σ_e; the wrong root is never closer than 7%.

Sequence. (i) Identify the H_2O^+ or N_2^+ ToF peak; read σ_m, the DCCT bunch charge N and the cage voltage V. (ii) On that species' table at (N, V) locate σ_0^* and $\sigma_{m,\min}$; if $\sigma_m \leq \sigma_{m,\min}$ within the width error, the beam is at the fold and $\sigma_0 = \sigma_0^*$. (iii) Otherwise return σ_S and σ_L. If σ_m exceeds the range of the small branch, $\sigma_S < 3$ mm is not an RCS beam and only σ_L remains; if it exceeds the large branch, $\sigma_L > 20$ mm lies beyond the table. (iv) Read σ_e from the electron mode at the same N and V, predict $T_e(\sigma_S)$ and $T_e(\sigma_L)$ from the $B = 0$ electron table, keep the root whose prediction matches, and confirm it with the other species; a root beyond the 20 mm end of the electron table cannot be checked and the other root is accepted only if its own prediction matches within 10%. (v) Check the detected fraction: sizes whose ions reach the 110 mm half-cage (fraction $< 99.5\%$) are not inverted; widths above 40 mm (within 2.75σ of the wall) are inverted but flagged, because the real cage map matters most there. Cage voltage and a BPM Δy enter as parameters of the same table ($\Delta \propto 1/V$ and $\sigma_m^2 - \sigma_0^2 \propto d$, Sections 4.2 and 4.3). The extraction column must be the aligned generation (Fig. 18): at 10 mm, 100 kW, 0 G, 25 kV the aligned expansions are $+104.1\%$ (H_2^+), $+40.8\%$ (H_2O^+) and $+40.0\%$ (N_2^+), not the v1 as-run H_2^+ $+33.5\%$. H_2^+ is not the working peak: at 100 kW its minimum sits at the design core itself (9 mm at injection, 10 mm at extraction), so its two roots merge exactly where the measurement is wanted.

The test below is leave-one-out: each tabulated σ_0 is removed, its σ_m is inverted on the remaining points, σ_e is the Virtual-IPM $B = 0$ electron width of the same beam, T_e is built from the electron table with the same point removed, and the residual is quoted against the true σ_0. The 3 and 20 mm grid ends (25 mm at extraction) are extrapolations under this test and are left out of the statistics. Figures 12 and 13 show each power separately; Table 5 follows the 10 mm design core and Table 6 the two branches.

4.4.1 Injection, 80 MeV

100 kW (panels a, e). The fold is at 8 mm for both species ($\sigma_{m,\min} = 16.01$ and 15.02 mm). The design core is on the large branch: $\sigma_m = 16.44$ / 15.60 mm gives $\sigma_S = 6.2$ / 5.6 mm and $\sigma_L = 10.05$ / 10.02 mm; the electron width of the same beam is $\sigma_e = 5.43$ mm (the -48% compression of Section 3.1), the roots predict $T_e(\sigma_S) = 8.4$ / 9.2 mm and $T_e(\sigma_L) = 5.6$ mm, so σ_L is the consistent

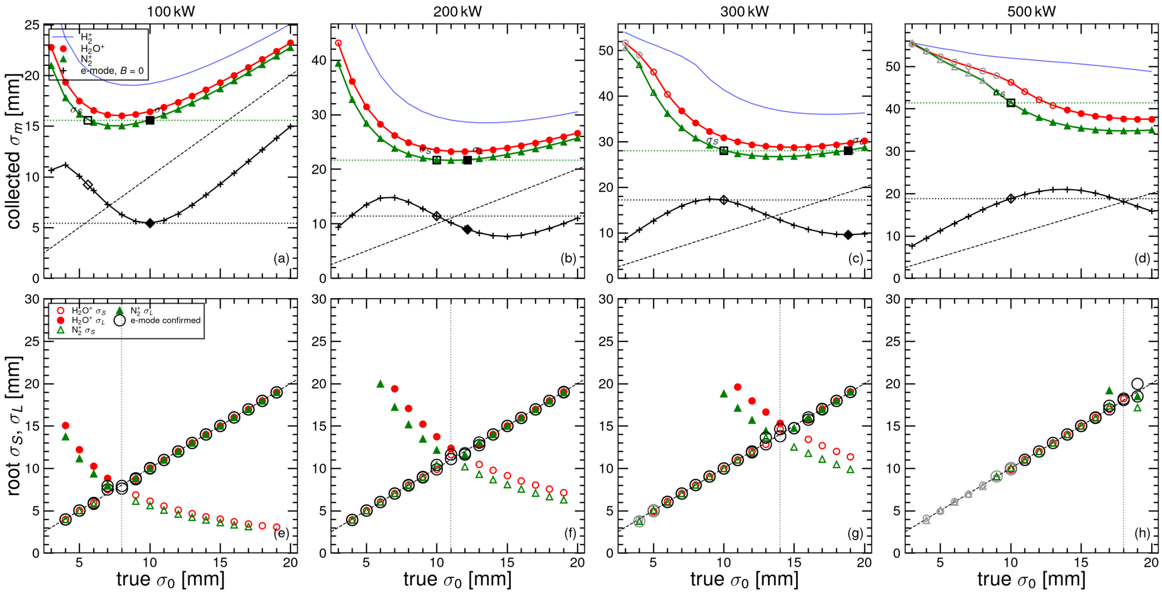

Fig. 12: Injection, 80 MeV, 25 kV, $B = 0$, Block B, one column per power. Top: collected σ_m of H_2O^+ (red) and N_2^+ (green) versus true σ_0; thin blue is H_2^+; the black curve is the electron width at $B = 0$; the dashed line is $\sigma_m = \sigma_0$. Open coloured markers are collected in full but above 40 mm; grey markers lose ions on the cage. The green dotted line is the N_2^+ width of a 10 mm beam with its two roots σ_S (open square) and σ_L (filled square); the black dotted line is the electron width of the same beam, and the diamonds are the electron widths the two roots predict, $T_e(\sigma_S)$ (open) and $T_e(\sigma_L)$ (filled)—the one on the dotted line is the consistent root. Bottom: leave-one-out roots σ_S (open) and σ_L (filled) for both species; the black ring marks the root confirmed by the electron width; the dotted vertical is the N_2^+ fold.

root (+0.5% / +0.2%; margin 52% / 66%). The slope $d\sigma_m/d\sigma_0$ at 10 mm is 0.36 / 0.42, so a 1% width error is a 2.4–2.8% size error on this branch. Painted beams 9–19 mm are recovered to a median 0.02% (maxima 3.0% and 1.8% at 9 mm, next to the fold). The small branch 4–7 mm recovers to 0.9% / 2.0% median but 7.6% / 13.5% at 7 mm, one step below the fold, where the width no longer changes with size. A wrong choice would be at least 15–24% off; the two species disagree on their wrong roots by 10%.

200 kW (b, f). The fold has moved to 12 / 11 mm (23.21 / 21.59 mm) and the 10 mm core is now the small root. $\sigma_m = 23.49$ / 21.70 mm, only 1.2% / 0.5% above the minimum, gives $\sigma_S = 9.89$ / 10.42 mm and $\sigma_L = 13.73$ / 12.19 mm. Here $\sigma_e = 11.44$ mm (+10%): $T_e(\sigma_S) = 11.6$ / 10.9 mm against $T_e(\sigma_L) = 7.9$ / 8.9 mm, so σ_S is consistent (−1.1% / +4.2%; margin 32% / 17%). The N_2^+ core sits one step below its fold (slope −0.23) and is ill-conditioned there, so σ_e carries the measurement and the N_2^+ width confirms $\sigma_0 \simeq \sigma_0^*$. Away from the fold the selected root is within 0.22% (median) and 1.3% (maximum) for both species: the core branch below the fold to 0.6%, the painted branch above it to 0.04%. Wrong roots are $\geq$ 12% off, and the two species disagree on them by 13%.

300 kW (c, g). Fold 15 / 14 mm (28.74 / 26.67 mm). The core is well inside the small branch: $\sigma_m =$ 30.89 / 28.07 mm, $\sigma_S = 10.02$ mm for both species (slope −1.14 / −0.90, better conditioned than at 100 kW), $\sigma_L > 20$ / 18.8 mm. The electron width is 17.20 mm (+65%); $T_e(\sigma_S) = 17.1$ mm matches it, while the N_2^+ painted root predicts 9.6 mm (margin 44%): σ_S, +0.2%. Sizes $\leq$ 4 mm (H_2O^+) and 3 mm (N_2^+) lose ions to the cage; 5–6 / 4–5 mm are collected in full but above 40 mm. The core branch recovers to 0.35% / 0.48%, the painted branch to 0.17% / 0.08%; within one step of the fold up to 4.8%.

500 kW (d, h). The fold has reached 19 / 18 mm: the whole operating range is the small branch and there is no painted-beam root inside the table. Sizes $\leq$ 9 / 8 mm lose 1.5–27% / 0.7–24% of their ions on the 110 mm half-cage and are not inverted. The 10 mm core is collected in full (99.9% / 100%) at $\sigma_m = 46.3$ / 41.4 mm, above the 40 mm flag; its small root is 9.90 / 10.05 mm (-1.0% / $+0.5\%$) with a slope of -1.9 / -2.3, the best conditioned of all powers; it is the only root, and the electron width confirms it ($\sigma_e = 18.79$ mm, $+80\%$; $T_e(\sigma_S) = 18.7$ / 18.8 mm). Inside the cage, the core branch from 10 mm up (9 mm for N_2^+) is recovered to 0.15% / 0.19% median (maxima 1.4% / 2.3%). With the fold at the grid end, 18–19 mm painted beams are read from σ_e. A single-branch invert on $\sigma_0 \geq \sigma_0^*$ has no inside-cage recovery at this power; the core hypothesis works down to 10 mm, provided the field-cage map near the electrodes is known.

4.4.2 Aligned extraction, 1.6 GeV

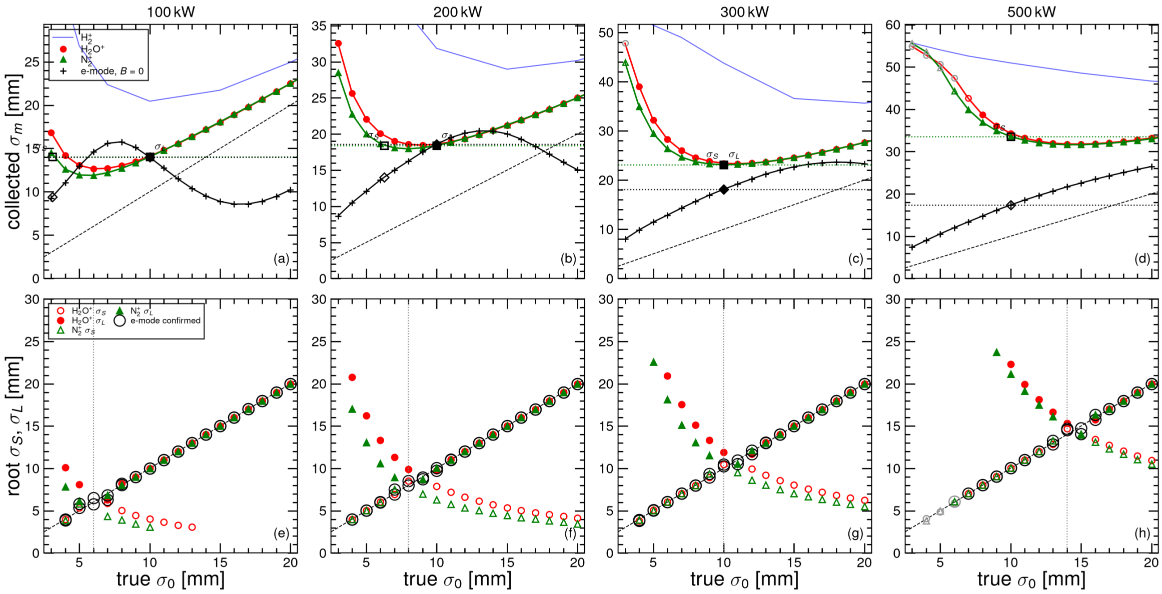

Fig. 13: As Fig. 12 for aligned extraction (1.6 GeV, $\beta = 0.929$, $\sigma_t = 20$ ns, generation centre $4\sigma_t$), 25 kV, $B = 0$, on the 1 mm I2 grid. The thin blue H_2^+ curve is the coarse I1 grid; the black electron curve is the $B = 0$ extraction electron table.

100 kW (a, e). Fold 6 / 6 mm (12.66 / 11.89 mm). The core is on the large branch with a steeper slope than at injection (0.65 / 0.73): $\sigma_m = 14.11$ / 14.03 mm, $\sigma_S = 4.1$ / 3.1 mm, $\sigma_L = 10.00$ / 10.00 mm, $\sigma_e = 13.99$ mm ($+33\%$) is matched by $T_e(\sigma_L) = 14.0$ mm and not by $T_e(\sigma_S) = 11.1$ / 9.4 mm: σ_L (-0.02%; margin 21% / 33%). Painted beams 7–20 mm: median 0.02% / 0.01%, maxima 9.9% (7 mm, at the fold) / 2.2%; 8–20 mm within 2.7%. The small branch is 4–5 mm, two points that are not RCS sizes. The wrong root is $\geq 16\%$ off and the two species disagree on it by 30%.

200 kW (b, f). Fold 9 / 8 mm (18.45 / 17.95 mm). $\sigma_m(10) = 18.59$ / 18.41 mm, $\sigma_S = 7.9$ / 6.3 mm, $\sigma_L = 9.72$ / 10.05 mm. $\sigma_e = 18.60$ mm ($+77\%$) against $T_e(\sigma_L) = 18.3$ / 18.7 mm and $T_e(\sigma_S) = 16.1$ / 14.0 mm: σ_L (-2.8% / $+0.5\%$; margin 12% / 25%). The H_2O^+ core is one step above its fold (slope 0.23); N_2^+ is the better-conditioned peak here. Painted branch to 0.02%; core branch to 0.7%; within one step of the fold up to 7.6%.

300 kW (c, g). Fold 11 / 10 mm (23.33 / 23.13 mm): the design core sits at the fold. N_2^+ has its minimum at 10 mm (slope -0.04) and returns the fold, 10.26 mm; H_2O^+ returns $\sigma_S = 10.50$ and $\sigma_L = 11.91$ mm; $\sigma_e = 18.13$ mm is nearer to $T_e(\sigma_S) = 18.7$ than to $T_e(\sigma_L) = 20.1$ mm (margin 8%): σ_S ($+5.0\%$). The ion width cannot resolve a 10 mm beam at this power; σ_e is the measurement and the ions confirm $\sigma_0 \simeq 10$–11 mm. Away from the fold the core branch recovers to 0.7% / 0.9% and the painted branch to 0.03%. The 3 mm H_2O^+ beam loses ions.

500 kW (d, h). Fold 15 / 14 mm (31.76 / 31.62 mm). The core is again well inside the small branch: $\sigma_m = 34.33$ / 33.55 mm, $\sigma_S = 10.03$ mm for both species (slope -1.5 / -1.1), $\sigma_L = 22.3$ / 21.1 mm. The painted root lies beyond the 20 mm end of the electron table and cannot be checked; σ_S is accepted because its own prediction, $T_e(\sigma_S) = 17.4$ mm, matches $\sigma_e = 17.37$ mm (+66%) to 0.3%: +0.3%. Sizes ≤ 6 / 5 mm lose ions (21% / 15% at 3 mm); 7 / 6 mm are collected above 40 mm. Core branch to 0.45% / 0.37%, painted branch to 0.06% / 0.09%; within one step of the fold up to 6.3%. This is the one case where the species cross-check is weak (wrong roots differ by only 3–4%): at 500 kW the H_2O^+ and N_2^+ curves nearly coincide, and the electron check carries the decision.

4.4.3 Selected root and design core

Figures 14–17 show, one power at a time, the residual of the confirmed root and the electron margin at both rings. The $B = 0$ electron width chose the right root at every one of the 247 inside-cage grid points (165 of them with two roots inside the tables; the rest have a single root, which the electron width confirms to within 10%). Away from the fold ($|\sigma_0 - \sigma_0^*| > 1$ mm, 200 points) the residual is 0.07% median; 92% of the points are within 1% and 97% within 2%, the painted branch to 0.02% and the core branch to 0.4% median. Within one step of the fold the two roots merge, the slope falls below about 0.25, and the leave-one-out residual rises to 3–8% (17% at the two-point 100 kW extraction fold); there σ_e is the measurement. These are interpolation errors of the table. A measured width adds $|\mathrm{d}\sigma_0/\mathrm{d}\sigma_m|$ times its own error: 1.4–2.8 at 100 kW, and below 1.1 on the core branch at 300–500 kW, where the small root is the best conditioned measurement of all (Table 5). Field-cage non-uniformity—J-PARC reported a factor-of-two shrink from the external map alone [11]—is not in the table and will have to be folded in once the CSNS measured map is available. Commissioning I2 powers (20–80, 150, 250 kW) stay on the 5 mm size grid and are not quoted here.

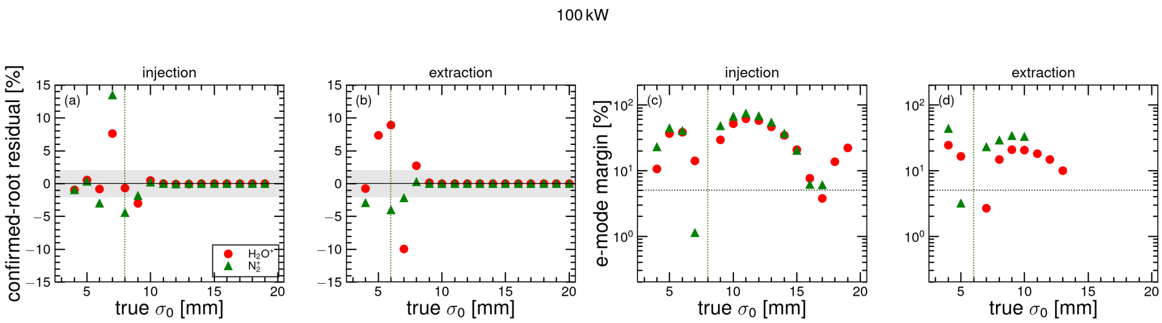

Fig. 14: 100 kW, $B = 0$, 25 kV. (a, b) Residual of the root confirmed by the $B = 0$ electron width versus true σ_0 (leave-one-out, grid interior, ions inside the cage) at injection and aligned extraction. Red circles: H_2O^+; green triangles: N_2^+; open markers are collected above 40 mm; the grey band is $\pm 2\%$; dotted verticals are the folds σ_0^* of the two species. (c, d) Electron margin $|T_e(\sigma_S) - T_e(\sigma_L)|/\sigma_e$ at the points with two roots inside the tables; the dotted horizontal is 5%.

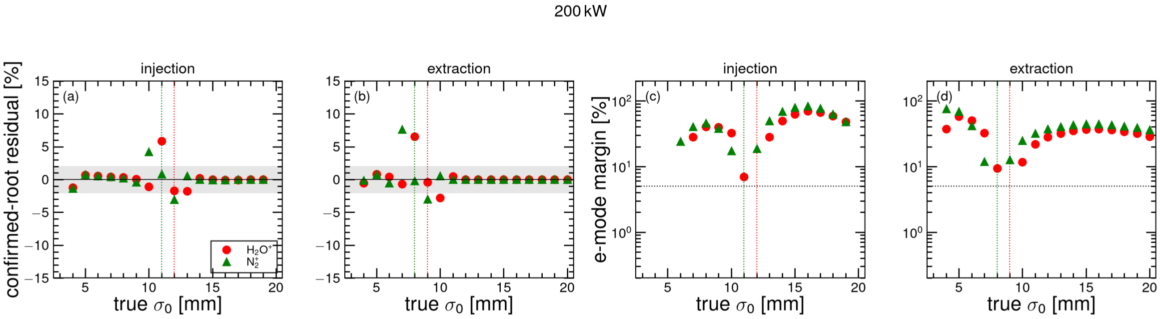

Fig. 15: As Fig. 14, at 200 kW.

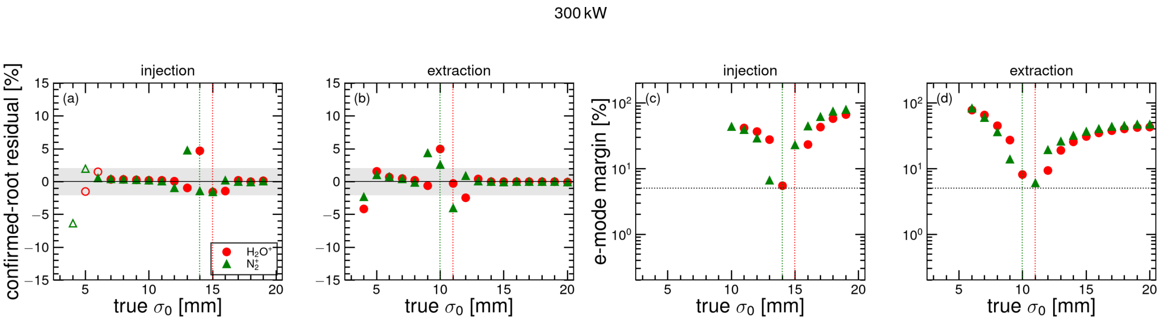

Fig. 16: As Fig. 14, at 300 kW.

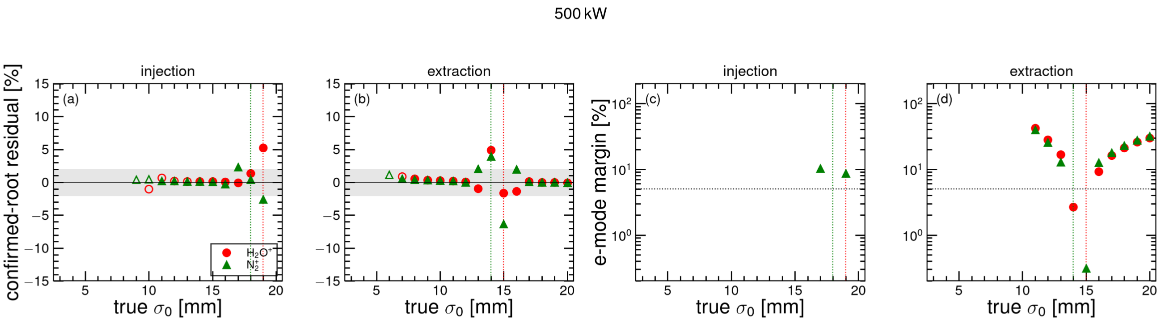

Fig. 17: As Fig. 14, at 500 kW.

Table 5: The $10\,\mathrm{mm}$ design core through the two-root invert, $B = 0$, $25\,\mathrm{kV}$ (injection Block B, extraction aligned I2, leave-one-out). σ_0^* locates the fold; $\sigma_m(10)$ and its slope are the tabulated ion width and $\mathrm{d}\sigma_m/\mathrm{d}\sigma_0$ at $10\,\mathrm{mm}$; σ_S, σ_L are the two roots; σ_e is the $B = 0$ electron width of the same beam and $T_e(\sigma_S)$, $T_e(\sigma_L)$ the electron widths the two roots predict; the last columns give the consistent root and its error. "> 20": the root lies beyond the ion table; ">tab.": beyond the electron table. "fold": the width equals the minimum.

Beam	P (kW)	peak	σ_0^* (mm)	$\sigma_m(10)$ (mm)	slope	σ_S (mm)	σ_L (mm)	σ_e (mm)	$T_e(\sigma_S)$ (mm)	$T_e(\sigma_L)$ (mm)	sel. (mm)	e (%)
Inj. 80 MeV	100	H_2O^+	8	16.44	$+0.36$	6.16	10.05	5.43	8.43	5.63	σ_L	$+0.5$
	100	N_2^+	8	15.60	$+0.42$	5.61	10.02	5.43	9.23	5.63	σ_L	$+0.2$
	200	H_2O^+	12	23.49	-0.37	9.89	13.73	11.44	11.60	7.90	σ_S	-1.1
	200	N_2^+	11	21.70	-0.23	10.42	12.19	11.44	10.90	8.94	σ_S	$+4.2$
	300	H_2O^+	15	30.89	-1.14	10.02	> 20	17.20	17.06	—	σ_S	$+0.2$
	300	N_2^+	14	28.07	-0.90	10.02	18.82	17.20	17.07	9.56	σ_S	$+0.2$
	500	H_2O^+	19	46.29‡	-1.88	9.90	> 20	18.79	18.65	—	σ_S	-1.0
	500	N_2^+	18	41.39‡	-2.29	10.05	> 20	18.79	18.82	—	σ_S	$+0.5$
Ext. 1.6 GeV	100	H_2O^+	6	14.11	$+0.65$	4.05	10.00	13.99	11.14	14.02	σ_L	-0.0
	100	N_2^+	6	14.03	$+0.73$	3.09	10.00	13.99	9.41	14.02	σ_L	-0.0
	200	H_2O^+	9	18.59	$+0.23$	7.91	9.72	18.60	16.14	18.31	σ_L	-2.8
	200	N_2^+	8	18.41	$+0.38$	6.28	10.05	18.60	14.04	18.65	σ_L	$+0.5$
	300	H_2O^+	11	23.42	-0.24	10.50	11.91	18.13	18.67	20.14	σ_S	$+5.0$
	300	N_2^+	10	23.13	-0.04	fold	fold	18.13	—	—	σ_0^*	$+2.6$
	500	H_2O^+	15	34.33	-1.46	10.03	22.29	17.37	17.42	>tab.	σ_S	$+0.3$
	500	N_2^+	14	33.55	-1.13	10.03	21.14	17.37	17.42	>tab.	σ_S	$+0.3$

‡Collected in full ($\geq 99.9\%$) but above $40\,\mathrm{mm}$, within 2.75σ of the $110\,\mathrm{mm}$ half-cage.

Table 6: Leave-one-out residual of the two branches, $B = 0$, $25\,\mathrm{kV}$, grid interior, ions inside the cage. For each branch: the sizes it covers, the median and maximum $|e|$ of the confirmed root. "fold": maximum $|e|$ within one step of σ_0^*. "pick": grid points at which the $B = 0$ electron width confirmed the correct root. "margin": smallest electron margin $|T_e(\sigma_S) - T_e(\sigma_L)|/\sigma_e$ on the grid. "lost": sizes whose ions reach the cage (not inverted). "> 40": sizes collected in full but above $40\,\mathrm{mm}$.

Beam	P	peak	core branch σ_S			painted branch σ_L			fold	pick	margin	lost	> 40
	(kW)		(mm)	med. (%)	max. (%)	(mm)	med. (%)	max. (%)	(%)		(%)	(mm)	(mm)
Inj. 80 MeV	100	H_2O^+	4–7	0.89	7.6	9–19	0.02	3.0	7.6	16/16	3.7	—	—
	100	N_2^+	4–7	1.97	13.5	9–19	0.02	1.8	13.5	16/16	1.1	—	—
	200	H_2O^+	4–11	0.62	5.9	13–19	0.04	1.8	5.9	16/16	7.0	—	3
	200	N_2^+	4–10	0.55	4.2	12–19	0.03	3.1	4.2	16/16	17.1	—	—
	300	H_2O^+	5–14	0.35	4.7	16–19	0.17	1.4	4.7	15/15	5.5	3–4	5–6
	300	N_2^+	4–13	0.48	6.3	15–19	0.08	1.6	4.8	16/16	6.6	3	4–5
	500	H_2O^+	10–18	0.15	1.4	—	—	—	5.3	10/10	—	3–9	10–13
	500	N_2^+	9–17	0.19	2.3	19	2.61	2.6	2.6	11/11	8.6	3–8	9–10
Ext. 1.6 GeV	100	H_2O^+	4–5	4.08	7.4	7–20	0.02	9.9	9.9	17/17	2.7	—	—
	100	N_2^+	4–5	9.92	16.9	7–20	0.01	2.2	16.9	17/17	3.2	—	—
	200	H_2O^+	4–8	0.70	6.5	10–20	0.02	2.8	6.5	17/17	9.4	—	—
	200	N_2^+	4–7	0.62	7.6	9–20	0.02	3.0	7.6	17/17	11.9	—	—
	300	H_2O^+	4–10	0.67	5.0	12–20	0.03	2.4	5.0	17/17	8.1	3	—
	300	N_2^+	4–9	0.90	4.4	11–20	0.03	4.0	4.4	17/17	6.0	—	3
	500	H_2O^+	7–14	0.45	4.9	16–20	0.06	1.4	4.9	14/14	2.6	3–6	7
	500	N_2^+	6–13	0.37	2.0	15–20	0.09	6.3	6.3	15/15	0.3	3–5	6

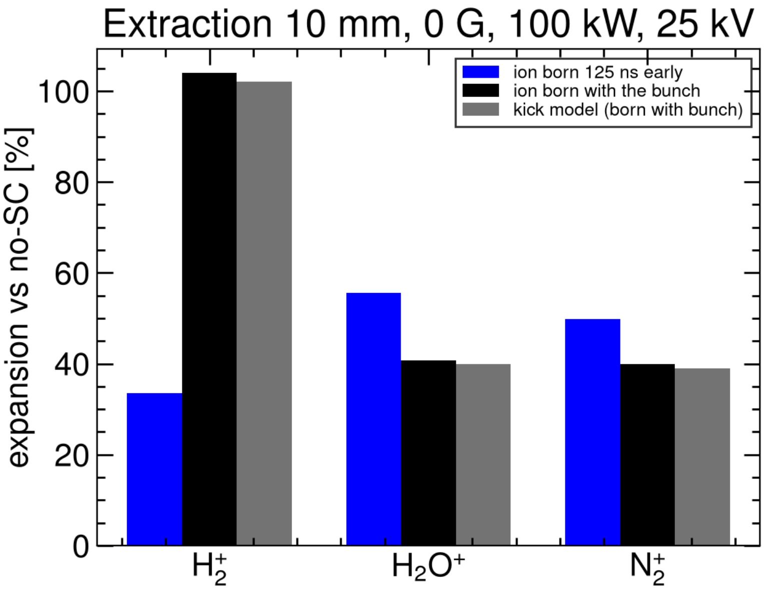

Fig. 18: Extraction, $10\,\mathrm{mm}$, $0\,\mathrm{G}$, $100\,\mathrm{kW}$, $25\,\mathrm{kV}$. Virtual-IPM with the ion born in time with the bunch matches the kick model. This is the extraction column of the inversion table.

4.5 Summary

Table 7 collects the efficiencies. Magnetic confinement at 300 G is the only mitigation that returns a faithful electron size. Cage voltage reduces the ion bias but does not bring an uncorrected 10 mm, 100 kW profile to the 10% level. An identified H_2O^+ or N_2^+ invert does, at injection and at aligned extraction, once the $B = 0$ electron width has chosen between its core and painted-beam roots and provided the ions stay inside the cage.

Table 7: Efficiency of the mitigations examined. Residuals are quoted for the families of Sections 3 and 4.

Mitigation	Electron mode	Ion mode
$B = 300\,\mathrm{G}$	tens of percent $\to \lesssim 1\%$ (all P, V, energy; $\sigma \geq 4\,\mathrm{mm}$)	—
Design $B = 0.1\,\mathrm{T}$	3 mm on the diagonal; residual $\leq 0.05\%$	—
Raise V ($5 \to 30\,\mathrm{kV}$)	not a size fix at $B = 0$; unused once $B \geq 300\,\mathrm{G}$	$+395\% \to +77\%$ (H_2^+, inj. 10 mm, 100 kW, $B = 0$); 10% would require $\sim 250\,\mathrm{kV}$
Orbit $\lesssim 10\,\mathrm{mm}$	300 G still $\lesssim 1\%$	Δx: none; Δy: a few percent
Invert $\sigma_m = \sigma_0\, h(\sigma_0, N, V)$	unnecessary at $\geq 300\,\mathrm{G}$	identify H_2O^+ or N_2^+; two roots σ_S (core) / σ_L (paint); the $B = 0$ electron width selects the root whose predicted electron width it matches (median margin 34%), the other species confirms; away from the fold 0.02% (paint) / 0.4% (core) median, 92% within 1%; within one step of the fold 3–8% and σ_e is used; 10 mm core $\leq 1\%$ at 100–500 kW except 200 kW N_2^+ inj. and 300 kW ext. (at the fold); lost on the cage: $\leq$ 9 mm at 500 kW inj., $\leq$ 6 mm at 500 kW ext.; H_2^+ is not used for the size

5 Conclusions

Space-charge distortion of the CSNS RCS IPM has been quantified with uniform-field tracking and checked against a reduced kick model and against the published electron-mode and ion-mode scalings.

Electrons experience an attractive kick that can focus or over-focus. The resulting $\Delta(B)$ is a cyclotron-phase oscillation whose envelope falls below 1% at 300 G for every beam, power, voltage and offset examined, and along the energy ramp. Size growth with charge, and the non-monotonic $\sigma_m(\sigma_0)$ at $B = 0$, both disappear at that field. The design 0.1 T is a margin, except for the smallest (3 mm) beams.

Ions experience a repulsive kick and are always collected with $\sigma_m > \sigma_0$. The inflation scales as N/σ_0^2 (impulsive) and as approximately $1/V$, and matches the kick model to 1%. Raising the cage voltage cannot bring a 10 mm, 100 kW ion profile to the 10% level. The width of one identified species (H_2O^+ or N_2^+) has a minimum at a size that grows with intensity, so its look-up inversion returns two candidates, a core size σ_S and a painted-beam size σ_L. With no guiding magnet yet, the $B = 0$ electron width chooses between them by consistency: each root predicts a $B = 0$ electron width from the electron table at the same bunch charge and voltage, the predictions differ by 34% in the median (below 5% only at the fold, where the roots coincide), and the root that matches the measured electron width is kept; the second ion species confirms the choice. Once the magnet is installed the electron width at $\geq 300\,\mathrm{G}$ is the size itself and the choice is immediate. Tested at 100, 200, 300 and $500\,\mathrm{kW}$ at both rings, the selected root is within 1% of the true size at 92% of the grid points (painted branch 0.02%, core branch 0.4% median), and the $10\,\mathrm{mm}$ design core is returned to $\leq 1\%$ at every power except within one step of the fold ($200\,\mathrm{kW}$ N_2^+ at injection, $300\,\mathrm{kW}$ at extraction), where the ion width is flat and the electron width is the measurement. At $500\,\mathrm{kW}$ the core branch is the best conditioned of all, but sizes below $10\,\mathrm{mm}$ at injection and $7\,\mathrm{mm}$ at extraction lose ions on the cage. H_2^+ is not used for the size. The extraction column is the aligned generation ($+104.1\%/+40.8\%/+40.0\%$ at $10\,\mathrm{mm}$), not the v1 as-run $+33.5\%$.

The recommended operating point is therefore electron collection at $B \gtrsim 300\,\mathrm{G}$. Ion time-of-flight identifies the residual gas; the beam profile is then taken from the H_2O^+ or N_2^+ peak after inversion, with the $B = 0$ electron width deciding between the core and the painted-beam root. Once space charge is removed, the remaining electron-mode systematic is hardware: microchannel-plate saturation after $\sim 200\,\mu\mathrm{s}$, electromagnetic interference, secondary electrons, and the real cage map [2, 3, 5].

Acknowledgements

The hardware parameters follow Ref. [4]. Tracking was performed with Virtual-IPM 2.3.1 [6]. This note is prepared for internal review.

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